

POS SYSTEM

Kenyan Supermarket

Complete Setup & User Guide

Version 1.0 — September 2026

NestJS + React + Electron + PostgreSQL

Table of Contents

1. System Requirements
2. Installation & Setup
3. Database Configuration
4. User Accounts & Login
5. Dashboard
6. POS Terminal (Barcode Scanning)
7. Products Management
8. Inventory Management
9. Sales & History
10. Customers & Loyalty
11. Cash Management
12. Reports & Analytics
13. Settings & Configuration
14. M-Pesa Integration
15. Barcode Scanner Setup
16. Troubleshooting
17. API Reference

1. System Requirements

Component	Requirement
Operating System	Windows 10/11, macOS, or Linux
Node.js	Version 18 or higher
PostgreSQL	Version 14 or higher (cloud or local)
RAM	Minimum 4 GB (8 GB recommended)
Display	1280×720 minimum (1920×1080 recommended)
Barcode Scanner	Any USB HID barcode scanner (keyboard-wedge type)
Receipt Printer	Optional — 80mm thermal printer
Internet	Required for M-Pesa payments, optional for offline use

2. Installation & Setup

2.1 Clone the Repository

```
git clone https://github.com/joshkirui/supermarket-pos.git
cd supermarket-pos
```

2.2 Install Backend Dependencies

```
npm install
```

2.3 Install Frontend Dependencies

```
cd frontend
npm install
cd ..
```

2.4 Build the Backend

```
npx nest build
```

2.5 Start the System

Open **two terminals** in the project folder:

Terminal	Command	What It Does
Backend	<code>node dist/main.js</code>	Starts API server on port 3001
Frontend	<code>cd frontend && npx vite --host</code>	Starts React app on port 5173

Tip: If PowerShell blocks scripts, run this first:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Open your browser to **<http://localhost:5173>**

3. Database Configuration

3.1 PostgreSQL Setup

Edit the `.env` file in the project root:

```
DATABASE_URL="postgresql://username:password@localhost:5432/pos_db"
JWT_SECRET="your-secret-key-here"
PORT=3001
```

3.2 Run Migrations

```
npx prisma migrate deploy
```

3.3 Seed the Database

```
node prisma/seed.js
```

This creates: permissions, roles, admin user, 12 products, payment methods, and categories.

3.4 Create Supermarket Accounts

```
node create-accounts.js
```

This creates all user accounts and sample customers listed in Section 4.

4. User Accounts & Login

4.1 Default Accounts

Username	PIN	Full Name	Role
admin	1234	Joshua Kirui	Super Admin
manager	5678	Faith Wanjiku	Branch Manager
cashier	1111	Daniel Ochieng	Cashier
cashier2	2222	Grace Muthoni	Cashier
inventory	3333	Peter Kamau	Inventory Clerk

4.2 Role Permissions

Permission	Super Admin	Manager	Cashier	Inventory
Create/View Sales	Yes	Yes	Yes	No
Refunds	Yes	Yes	No	No
Process Payments	Yes	Yes	Yes	No
View/Manage Products	Yes	Yes	View only	Yes
Inventory Adjust/Transfer	Yes	Yes	No	Yes
Customer Management	Yes	Yes	Create/View	No
View Reports	Yes	Yes	No	No
Cash Session	Yes	Yes	Open/View	No
User Management	Yes	No	No	No

4.3 Logging In

1. Open **http://localhost:5173**
2. Enter your **username** and **4-digit PIN**
3. Click **Login**
4. You will be redirected to the Dashboard

Security: After 3 failed login attempts, the account may be locked.
Contact the admin to reset.

5. Dashboard

The Dashboard shows real-time business metrics:

Key Performance Indicators (KPIs)

- **Today's Sales** — Total revenue for today in KES
- **Transactions** — Number of sales completed today
- **Average Sale** — Average transaction value
- **Low Stock Items** — Products below reorder level

Charts

- **Weekly Sales** — Bar chart showing daily sales for the week
- **Recent Sales** — List of the latest transactions

6. POS Terminal (Barcode Scanning)

This is the main screen for processing sales. Navigate to **Sales** → **POS Terminal**.

6.1 Scanning Items

1. The **barcode input** is always focused — no need to click it
2. Point your barcode scanner at a product barcode
3. The scanner types the barcode number and presses Enter automatically
4. The product appears in the cart table below
5. Product name, SKU, price, and quantity are shown

6.2 Manual Entry

1. Click the barcode input field
2. Type the barcode number manually
3. Press 

6.3 Adjusting Quantities

- Click **+** to increase quantity
- Click **–** to decrease quantity
- Click the **trash icon** to remove an item

6.4 Completing a Sale

1. Check the **Order Summary** panel on the right
2. Verify Items count, Subtotal, VAT (16%), and Total
3. Click **Complete Sale**
4. Select payment method (Cash / M-Pesa / Card)
5. Confirm payment
6. Receipt is generated and saved

VAT: All prices include 16% VAT as required by Kenya Revenue Authority (KRA).

7. Products Management

Navigate to **Products** in the sidebar.

7.1 View Products

- Products displayed as cards with name, SKU, category, price, and stock level
- Use the **search bar** to filter by name, SKU, or barcode
- **Click any product** to add it to the current cart

7.2 Add a New Product

1. Click **Add Product** button (top right)
2. Fill in the form:
 - **SKU *** — Unique stock keeping unit code (e.g., `MILK-003`)
 - **Product Name *** — Full product name (e.g., `Fresh Milk 1L`)
 - **Barcode** — EAN-13 barcode number
 - **Selling Price** — Price in KES
 - **Cost Price** — Your purchase cost in KES
 - **Initial Stock** — Starting quantity
 - **Reorder Level** — Minimum stock before alert (default: 10)
3. Click **Add Product**

7.3 Product Categories

Category	Example Products
Dairy & Eggs	Milk, Eggs, Yogurt
Beverages	Water, Soda, Juice
Snacks	Chips, Biscuits, Nuts
Household	Rice, Sugar, Cooking Oil

Personal Care	Soap, Tissues, Shampoo
Bakery	Bread, Cakes

8. Inventory Management

Navigate to **Inventory** in the sidebar.

8.1 View Stock Levels

- **All Stock** tab — shows all products with current stock levels
- **Low Stock** tab — shows only products below reorder level (amber alerts)
- Search by product name or SKU

8.2 Adjust Stock

1. Click **Adjust Stock** button
2. Select the product from the dropdown
3. Choose adjustment type:
 - **Restock** — New delivery from supplier
 - **Adjustment** — Stock count correction
 - **Damage** — Damaged/expired items
 - **Return** — Customer returns
4. Enter the quantity
5. Add a reason (e.g., "Supplier delivery - invoice #1234")
6. Click **Adjust**

8.3 Transfer Stock

1. Click **Transfer** button
2. Select the product
3. Enter the quantity to transfer
4. Set the **From Terminal** and **To Terminal** IDs
5. Add a reason (e.g., "Inter-branch transfer")
6. Click **Transfer**

9. Sales & History

Navigate to **Sales** in the sidebar. Two tabs:

9.1 POS Terminal

See Section 6 above for full barcode scanning workflow.

9.2 Sales History

- View all completed transactions
- Search by sale number (e.g., `SL-20260902-001`)
- See sale date, cashier, payment method, and total
- Click a sale to view details or print receipt

10. Customers & Loyalty

Navigate to **Customers** in the sidebar.

10.1 View Customers

- Table shows Name, Phone, Email, and Loyalty Points
- Search by name, phone, or email

10.2 Add a Customer

1. Click **Add Customer**
2. Enter Phone Number (required — used as unique ID)
3. Enter Name and Email (optional)
4. Customer is saved and can earn loyalty points on future purchases

10.3 Loyalty Program

- Customers earn **1 point per KES 1** spent (configurable)
- Points are auto-calculated on each sale
- Tier upgrades: **Bronze** → **Silver** (KES 10,000+) → **Gold** (KES 50,000+)
- Points can be redeemed at checkout to discount the total

11. Cash Management

Navigate to **Cash** in the sidebar.

11.1 Cash Session

1. **Open Session** — Enter the opening float amount (e.g., KES 10,000) and start your shift
2. **Cash In** — Record money added to the register (e.g., float top-up)
3. **Cash Out** — Record money removed (e.g., bank deposit)

11.2 Expenses

1. Switch to **Expenses** tab
2. Select category: Utilities / Supplies / Maintenance / Other
3. Enter amount and description
4. Click **Record Expense**

12. Reports & Analytics

Navigate to **Reports** in the sidebar.

12.1 Summary Cards

- **Total Sales (Week)** — Weekly revenue with % change from last week
- **Total Expenses** — Weekly expenses as % of sales
- **Net Profit** — Revenue minus expenses with margin %

12.2 Charts

- **Sales vs Expenses** — Daily comparison bar chart
- **Payment Methods** — Pie chart showing Cash/M-Pesa/Card split

12.3 Top Products

Table showing best-selling products by units sold and revenue.

13. Settings & Configuration

Navigate to **Settings** in the sidebar.

13.1 Business Info

- Business Name, Phone, Address
- KRA PIN (required for eTIMS compliance)

13.2 M-Pesa Settings

- Consumer Key & Secret (from Safaricom Developer Portal)
- Shortcode (Paybill or Till number)
- Passkey
- Callback URL for payment confirmations

13.3 Terminal Settings

- Terminal ID (e.g., `T-001`)
- Branch name
- VAT Rate (default: 16%)

13.4 Notifications

- Low stock alerts
- Daily sales summary
- Email notifications (optional)

14. M-Pesa Integration (STK Push)

14.1 How It Works

1. Cashier selects **M-Pesa** as payment method at checkout
2. Customer enters their M-Pesa phone number
3. System sends an **STK Push** to the customer's phone
4. Customer enters their M-Pesa PIN on their phone
5. Payment callback confirms success or failure
6. Sale is completed and receipt generated

14.2 Setup Steps

1. Register at <https://developer.safaricom.co.ke>
2. Create an app and get Consumer Key + Secret
3. Set up a Paybill or Till Number
4. Configure the callback URL in both Safaricom portal and POS Settings
5. Enter all credentials in **Settings** → **M-Pesa Settings**

Note: M-Pesa requires a Safaricom Lipa Na M-Pesa sandbox for testing. Use test credentials before going live.

15. Barcode Scanner Setup

15.1 Compatible Scanners

Any USB barcode scanner that works as a **keyboard input device** (HID mode) is compatible. Popular models:

- Honeywell Voyager 1200g
- Zebra LS2208
- Datalogic QW2520
- Any affordable 1D barcode scanner from Jumia/Kilimall

15.2 Setup

1. Plug the scanner into a USB port
2. The scanner should be in **keyboard/wedge mode** (default for most scanners)
3. Open the POS Terminal (Sales page)
4. The barcode input is **always focused** — just scan directly
5. The scanner reads the barcode, types the numbers, and presses Enter
6. The product is automatically added to the cart

15.3 Testing Your Scanner

Open Notepad, scan a barcode. If numbers appear, the scanner is working correctly.

15.4 Barcode Format

- The system uses **EAN-13** barcodes (13 digits)
- Sample barcodes in the seed data:
 - `6161100100011` — Fresh Milk 500ml

- 6161100100028 — Fresh Milk 1L
- 6161100300028 — Coca-Cola 500ml
- 6161100600011 — White Bread 400g

16. Troubleshooting

Problem	Solution
Blank white screen	Clear browser localStorage, refresh. Make sure Vite is running.
"Invalid credentials" on login	Check username/PIN. Default: <code>admin</code> / <code>1234</code> . Ensure backend is running on port 3001.
"No product found" when scanning	Check if the barcode is in the database. Add product with matching barcode.
Crashes on page navigation	Refresh the page. Check browser console for errors.
Backend won't start (port in use)	Run <code>netstat -ano findstr :3001</code> , kill the process, restart.
PowerShell blocks <code>npx</code>	Run <code>Set-ExecutionPolicy -Scope Process - ExecutionPolicy Bypass</code> first.
Database connection error	Check <code>.env</code> file. Ensure PostgreSQL is running. Verify username/password.
"Argument skip is missing" error	Ensure backend is rebuilt: <code>npx nest build</code>
Scanner not adding items	Ensure POS Terminal tab is active. Check if the scanner types into Notepad.
M-Pesa payment not going through	Check M-Pesa credentials in Settings. Ensure callback URL is accessible.

17. API Reference

Base URL

http://localhost:3001/api

Authentication

All endpoints require a JWT Bearer token in the `Authorization` header.

Core Endpoints

Method	Endpoint	Description
Auth		
POST	/auth/login	Login with username + PIN
Products		
GET	/products	List all products
POST	/products	Create a product
PUT	/products/:id	Update a product
GET	/products/scan/:barcode	Look up product by barcode
POST	/products/:id/variants	Add a variant to product
Sales		
GET	/sales	List sales
POST	/sales	Create a sale
GET	/sales/:id	Get sale details
GET	/sales/:id/receipt	Download PDF receipt

Customers		
GET	/customers	List customers
POST	/customers	Create/upsert customer
GET	/customers/phone/:phone	Find by phone
Inventory		
GET	/inventory	List stock levels
GET	/inventory/low-stock	Low stock alerts
POST	/inventory/adjust	Adjust stock level
POST	/inventory/transfer	Transfer stock
Payments		
POST	/payments/initiate	Initiate M-Pesa STK Push
GET	/payments/:id/status	Check payment status
Cash Management		
POST	/cash-sessions/open	Open cash session
POST	/cash-sessions/:id/close	Close cash session
GET	/cash-sessions/:id/summary	Session summary
Reports		
GET	/reports/daily-summary	Daily sales summary
GET	/reports/top-products	Top selling products
GET	/dashboard/kpis	Dashboard metrics
Settings		
GET	/settings	Get all settings
POST	/settings	Update setting

Swagger Documentation

Interactive API docs available at: **<http://localhost:3001/docs>**

POS System — Kenyan Supermarket
Built with NestJS, React, Tailwind CSS, Prisma, PostgreSQL
© 2026 — All rights reserved